

Lecture 5

Elastic blade characteristics

ASDA/Dynamics of Rotor Systems

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QB 2.64 UoB

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This lecture

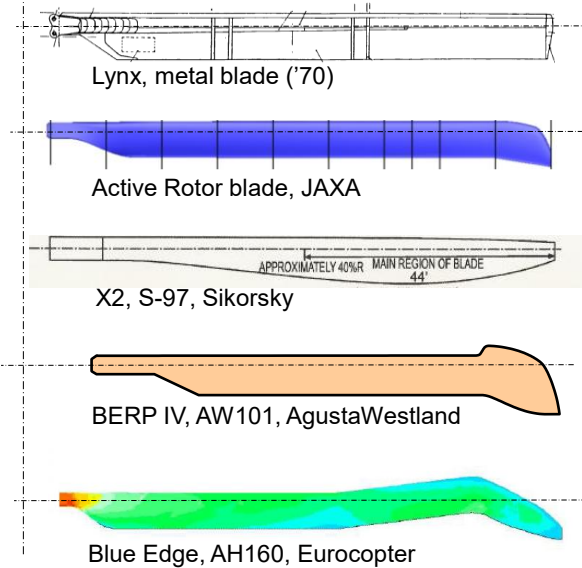
- Blade modelling
- 1D continuum representation
- Blade loads and motions
- Blade parameters
- Blade definition tables

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Diverse range of blade platforms and planforms

Historically, rotor blades evolved from wood-based and straight lift-generating components toward all-composite non-straight structures.

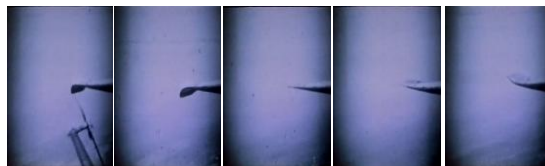


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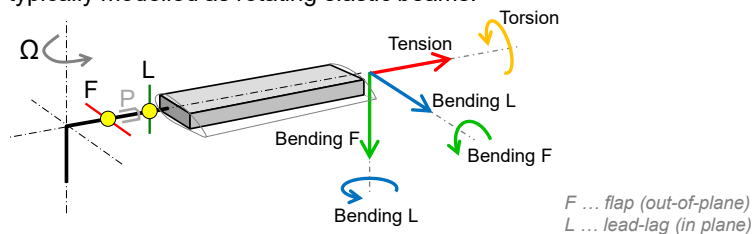
Blade motions and loads

Blades are *high aspect ratio rotating lifting surfaces* subject to range of varying loads:



This series of snapshots from a camera placed on the rotor hub illustrates the extent of the blade's dynamic response during its rotation.

Blades are typically modelled as rotating elastic beams:



Each loading scenario (bending, tension, torsion) requires knowledge of the dedicated cross-sectional characteristics and reference points: **shear center s.c.** (bending), tension center t.c. (tension), center of twist (twist). Typically, these reference points are chosen such that the corresponding forces applied in these points produce decoupled blade responses.

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Blade sectional properties

Blade description:

- Blades are slender flexible structures which are typically modelled as **beams**.
- The required blade sectional characteristics are closely associated with their dynamics, elasticity and aerodynamics properties.



Given anticipated strong *aeroelastic challenges* involved in normal blade operation, all three interacting disciplines pose requirements on the complete model description.

Examples



AH-64 (reddit.com)



BO105 (SABRE)

Additionally, blade sectional arrangements come in a variety of configurations arising from their typical operational requirements (e.g., redundancy, cost, performance, etc.)

Aerodynamics

Typical sectional properties:
aerodynamic centre (a.c.),
aerodynamic coefficients C_L , C_D ,
 C_M

Note: typically, quasi-steady
aerodynamics, strip theory (2D
aerodynamic blade stations).

Dynamics

Typical sectional properties:
centre of mass (c.m.), mass
distribution, inertial axes, polar
moment of inertia, ...

Elasticity

Euler-Bernoulli's beam bending
St. Venant's bar torsion

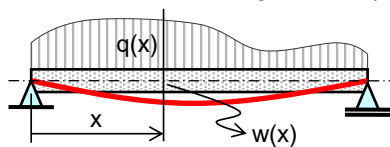
See next ...

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Recall simple elastic beam bending

Example simply supported beam configuration (no rotation):



$$M(x) = EI(x) \frac{d^2 w(x)}{dx^2}$$

This expression represents the Euler-Bernoulli's constitutive law which creates a link between the deformation geometry and associated internal loading - from the strains to the stresses and sectional loads.

Euler-Bernoulli beam theory assumptions:

- 1D representation of slender 3D structures,
- linear isotropic elasticity described by Hooke's law ($\sigma = E\epsilon$)
- section remains plane and normal to the elastic axis.

Other assumptions:

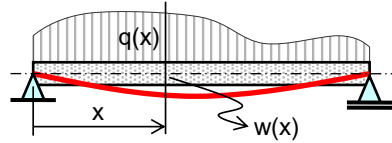
- small deformations $w(x)$ and rotations (geometrically linear),
- decoupled flap and lag (symmetric bending), etc.

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Recall simple elastic beam bending

Example simply supported beam configuration (no rotation):



$$M(x) = EI(x) w''(x)$$

Understanding the links between the internal sectional beam loads (from *Structures*), we can use the *constitutive relationship* to write the equation of elastic equilibrium – to be used to find the deflection state $w(x)$, etc.

Relationships between internal and external loads:

$$M'(x) = -S(x)$$

M ... internal bending moment

$$S'(x) = -q(x)$$

S ... internal shear force

q ... external distributed load

Differential bending equation:

$$(EI(x) w''(x))'' = q(x)$$

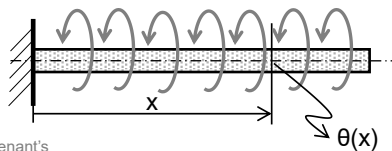
This equation can be solved (*integrated*) to find the $w(x)$ and, from that, further quantities can be found such as $M(x)$, etc.

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Recall simple elastic bar torsion

Example clamped elastic bar experiencing torsion (no rotation):



This expression represents the St. Venant's constitutive law which creates a link between the rotational deformation geometry and associated internal loading - from the strains to the stresses and sectional torques.

$$M = GJ \theta'$$

GJ (J [m⁴] ... polar moment of area) for solid circular section
 GK_T (K_T [m⁴] ... torsional constant) for thin open/closed wall sections
M ... internal torsional moment

Constitutive relationship:

St. Venant's model assumptions:

- sections does not change in-plane shape,
- sections rotate about the center of twist,
- out-of-section deformation (warping) is permitted.

Differential torsion equation:

$$(GJ(x) \theta'(x))' = -t(x)$$

$t(x)$ is a distributed applied torque

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Frequently used sectional parameters

Combined parameters:

- EI ... bending stiffness distribution (flexural rigidity) $\rightarrow EI_F, EI_L$
- GJ ... torsional stiffness distribution (or GK_T)
- ρA ... mass distribution ($\rho A = m$)
- ρJ ... mass moment of inertia distribution
- EA ... axial stiffness distribution

Basic parameters:

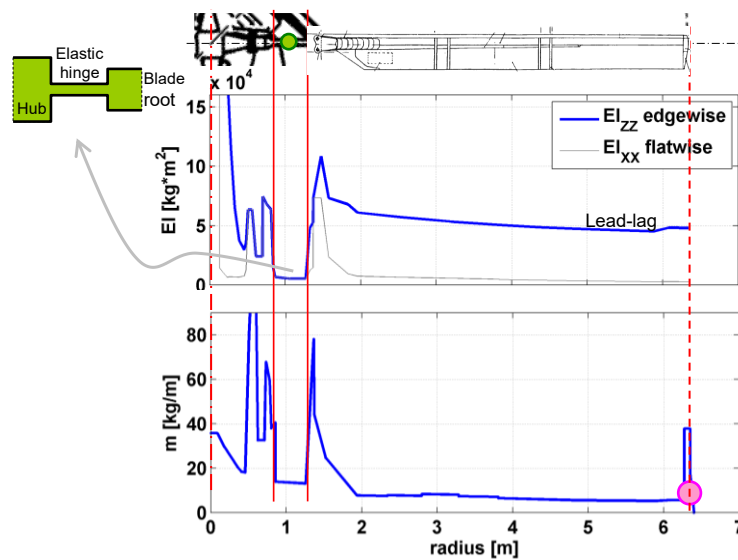
- ρ ... density, $[\text{kg}/\text{m}^3]$
- E ... Young's modulus (modulus of elasticity), $[\text{Pa}]$
- G ... shear modulus (modulus of rigidity), $[\text{Pa}]$
- A ... sectional area, $[\text{m}^2]$
- I ... second moment of area, $[\text{m}^4]$
- J ... polar moment of inertia, $[\text{m}^4]$
- K_T ... torsional constant K_T (general section), $[\text{m}^4]$
- e ... distance between e.c. and c.m., $[\text{m}]$
- ...

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Example: distributed $EI(x)$ and $m(x)$

Lau et al., *Performance and Rotor Loads Measurements of the Lynx XZ170 Helicopter with Rectangular Blades*, NASA Technical Memorandum 104000



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Example: detailed blade definition tables

Jung, Lau, *Determination of HART I Blade Structural Properties by Laboratory Testing*, NASA/CR-2012-216039 (page 25)



TABLE 14. SUMMARY OF HART I BLADE PROPERTIES

r/R	Y_{EA} , m	Z_{EA} , m	Y_{CG} , m	Z_{CG} , m	Twist deg	Mass kg/m	EI_{YAP} , N-m ²	EI_{ZAP} , N-m ²	GJ, N-m ²	k_{θ} , m	$I_{p,CG}$, kg-m	$I_{p,CG}$, kg-m
0	0	0	0	0.0001	3.84	10.706	25482.82	25482.82	10000.65	0.0056	0.000356	0.000311
0.025	0	0	0	0.0001	3.84	10.706	25482.82	25482.82	10000.65	0.0056	0.000356	0.000311
0.0251	0	0	0	0.0001	3.84	10.529	41237.32	41237.32	10000.65	0.0056	0.000356	0.000311
0.05	0	0	0	0.0001	3.84	10.529	41237.32	41237.32	10000.65	0.0056	0.000356	0.000311
0.0501	0	0	0	0.0001	3.84	9.633	39486.87	61984.91	10000.65	0.0192	0.003559	0.003514
0.0699	0	0	0	0.0001	3.84	9.633	39486.87	61984.91	10000.65	0.0192	0.003559	0.003514
0.07	0	0	0	0.0001	3.84	10.706	1004.57	6600.15	438.03	0.0266	0.007562	0.007517
0.095	0	0	0	0.0001	3.84	10.706	1004.57	6600.15	438.03	0.0266	0.007562	0.007517
0.0951	0	0	0	0.0001	3.84	1.321	656.1	5767.9	166.45	0.0124	0.000222	0.000178
0.12	0	0	0	0	3.84	1.321	656.1	5767.9	166.45	0.013	0.000222	0.000178
0.1201	0	0	0	0	3.84	1.25	528.3	4961.69	146.37	0.013	0.000222	0.000178

Other examples:

- Saito et al., Overview of the Novel Intelligent JAXA Active Rotor Program, Heli Japan 2010, AHS Int. Meeting on Advanced Rotorcraft Technology and Safety Operations, Ohmiya, Japan, November 1–3, 2010
- Jonkman et al., Definition of a 5-MW Reference Wind Turbine for Offshore System Development, National Renewable Energy Laboratory Technical Report NREL/TP-500-38060, 2009

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Summary

- Blade modelled as 1D beam continuum
 - The assumed modes of response and loading include bending, torsion, extension.
- Depending on the internal structure and material complexity, appropriate beam theories are chosen to describe beam elastic behaviour. Simplest examples include:
 - Bending: Euler-Bernoulli beam bending theory.
 - Torsion: St. Venant bar torsion theory.
 - (Tension: *not considered*)
- Blade sectional properties are characterised by the sectional parameters which arise from all three interacting disciplines:
 - EI , GJ , EA , ρA , ρJ , e , C_I , ...

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